PROJECT TECHNICAL OVERVIEW | MAINTAINER: SAMARTH MISHRA

MediSetu: Full-Stack Health Reference & Campus Notice Platform

Bilingual Lab & Prescription Shorthand Reader, Sleep-Debt Assessment, First-Aid Guide & Session Board

1. Problem Definition

Students, parents, and elderly household members routinely receive blood test sheets and prescriptions written in Latin abbreviations (`Hb, 25-OH Vit D3, OD BBF, BD PC, SOS, HS`) without plain-language explanations or supplement-food timing warnings (for example, milk calcium reducing iron absorption). At the same time, exam-year students lack straightforward sleep-debt checks and verified first-aid reference steps.

2. Full-Stack Architecture & API Endpoints

- **`POST /api/analyze` (Lab & Prescription Reader):** Evaluates laboratory biomarkers against standard reference intervals, decodes nine common prescription abbreviations into English and Hindi, flags Iron and Calcium timing conflicts, and pairs with browser speech synthesis for audio playback.
- **`POST /api/burnout` (Sleep & Fatigue Index):** Computes a readiness score from sleep hours, daily screen/study load, stress level, and hydration alongside a 4-4-4-4 breathing pacer and India **14416 Tele-MANAS** helpline details.
- **`GET & POST /api/community` (Campus Notice Board):** Supports sanitized session notices for study groups, health checkup camps, and hydration stations.
- **Legal, Privacy & Accessibility Pages:** Includes a vector SVG favicon (`/favicon.svg`), dedicated Privacy Policy (`/privacy.html`, with one-click local storage clearing) and Terms & Conditions (`/terms.html`) pages, flat neutral-slate and emerald typography, and inline geometric icons.

3. Stack & Automated Verification

Built with **HTML5, CSS3, Vanilla ES6+ JavaScript, Node.js HTTP / Vercel Serverless Functions, Browser Web Speech Synthesis, and Automated Static + End-to-End QA (`npm test`)** with zero external tracking dependencies.